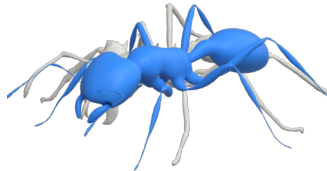
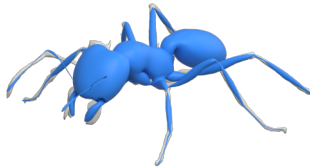
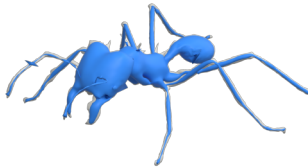
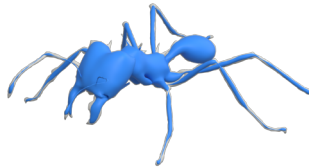


**a** init**b** pose**c** Stage 2 (coarse)**d** Stage 3 (fine)

chamfer  $1.35 \times 10^{-2}$   
 $F_{0.01}$  0.149  
penetration 18



chamfer  $8.88 \times 10^{-4}$   
 $F_{0.01}$  0.508  
penetration 46



chamfer  $8.30 \times 10^{-5}$   
 $F_{0.01}$  0.929  
penetration 3



chamfer  $7.02 \times 10^{-5}$   
 $F_{0.01}$  0.947  
penetration 6

 fitted template

 scan surface not covered by the fit